

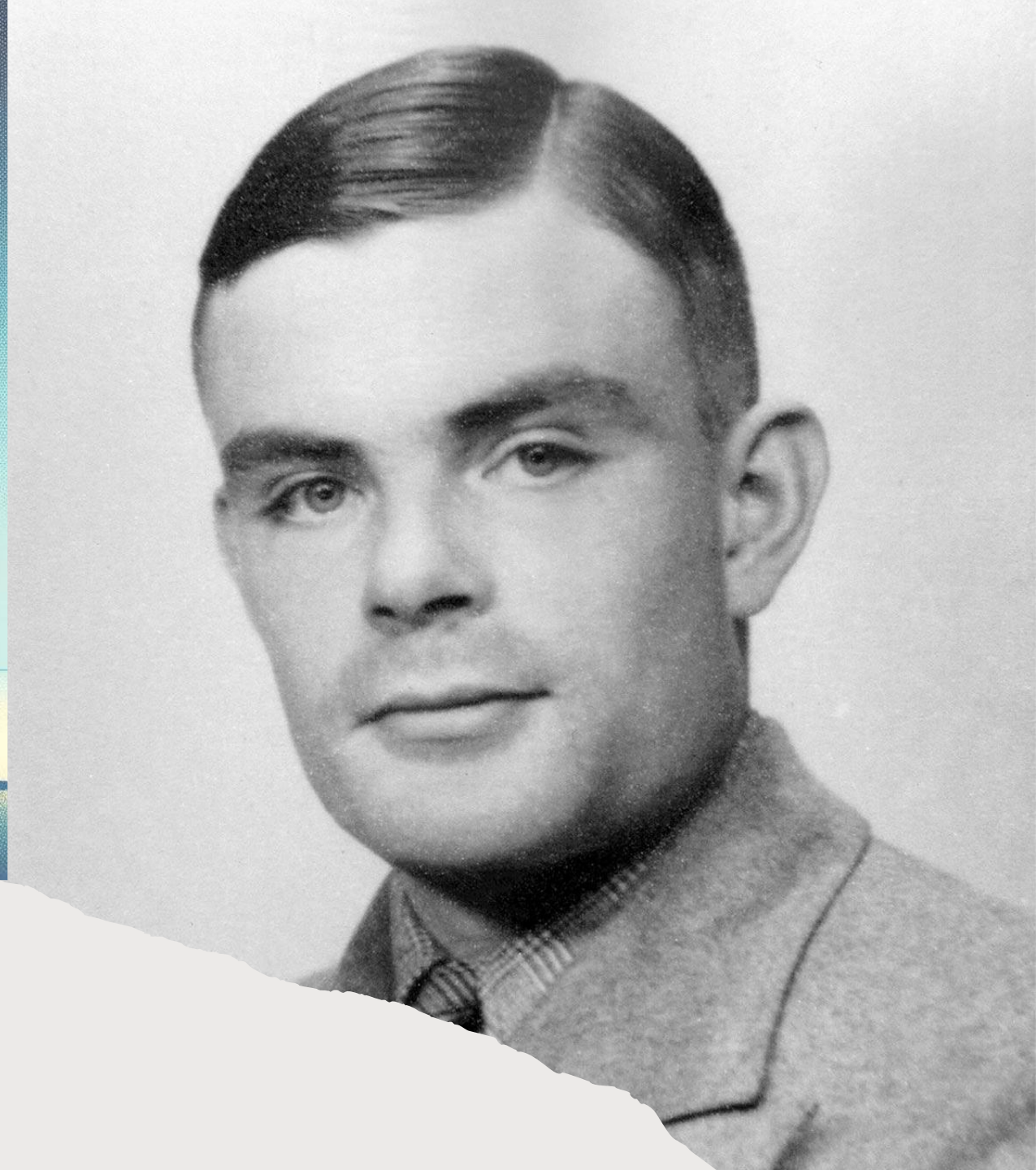
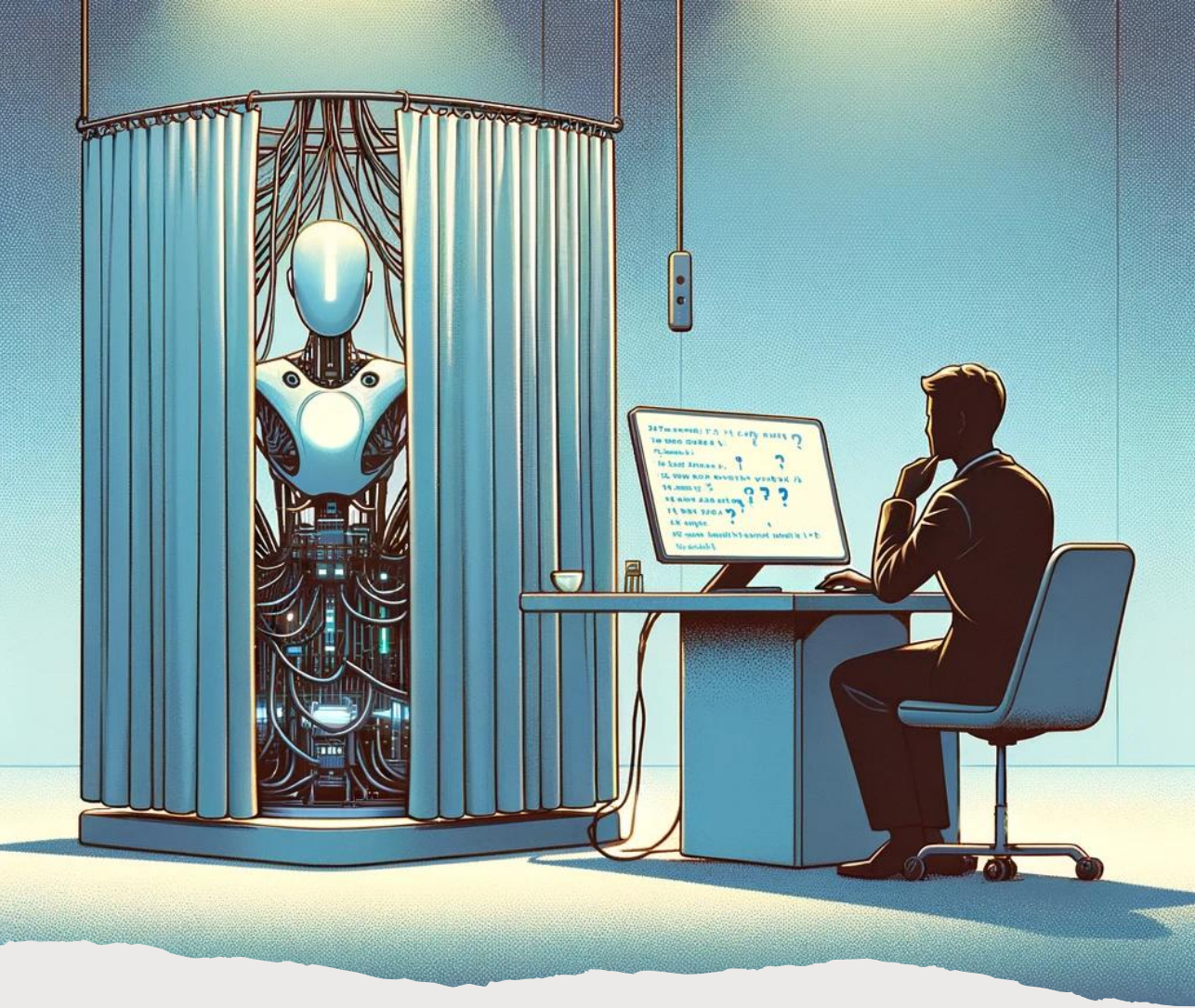
Introduction to Artificial Intelligence



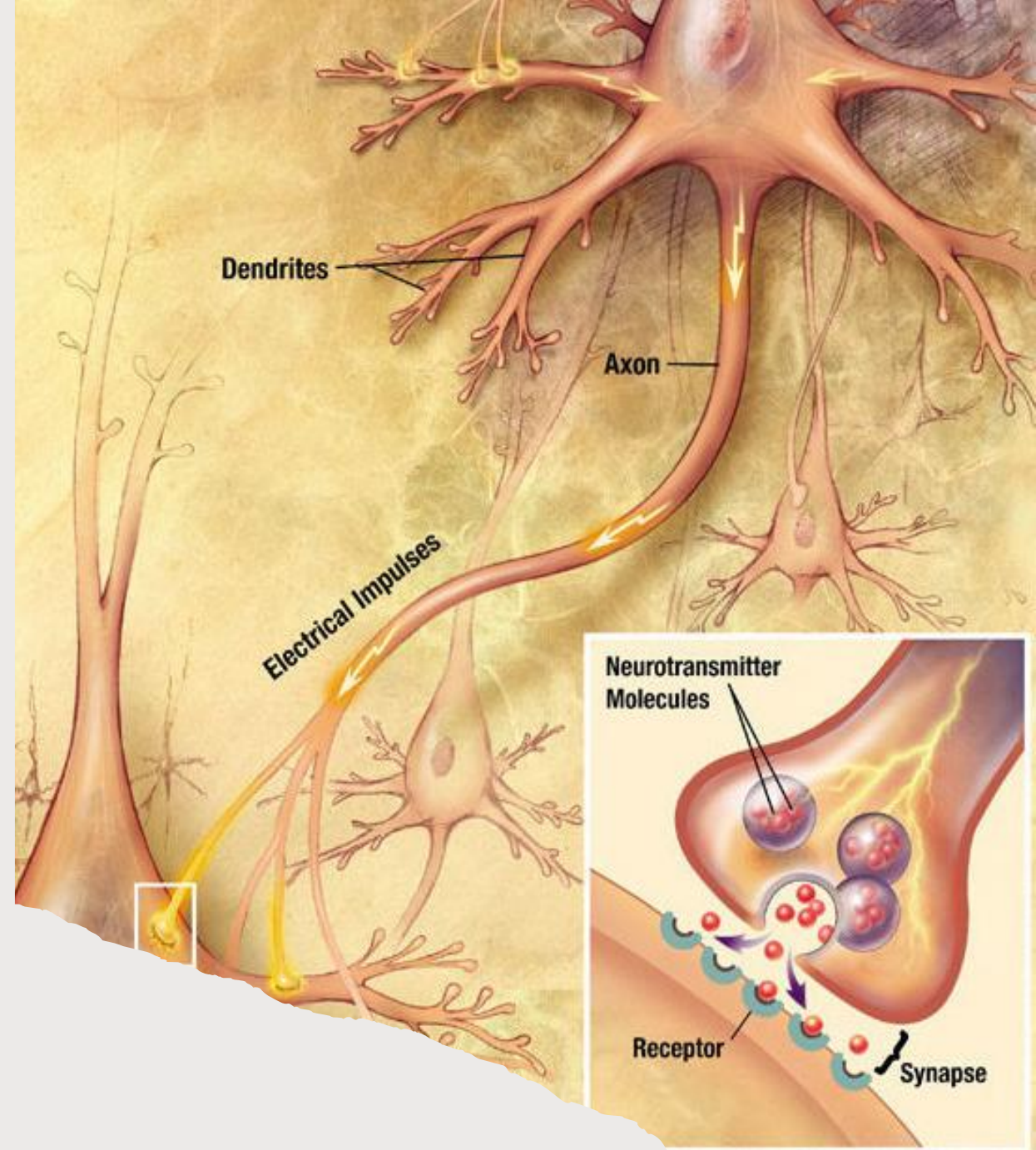
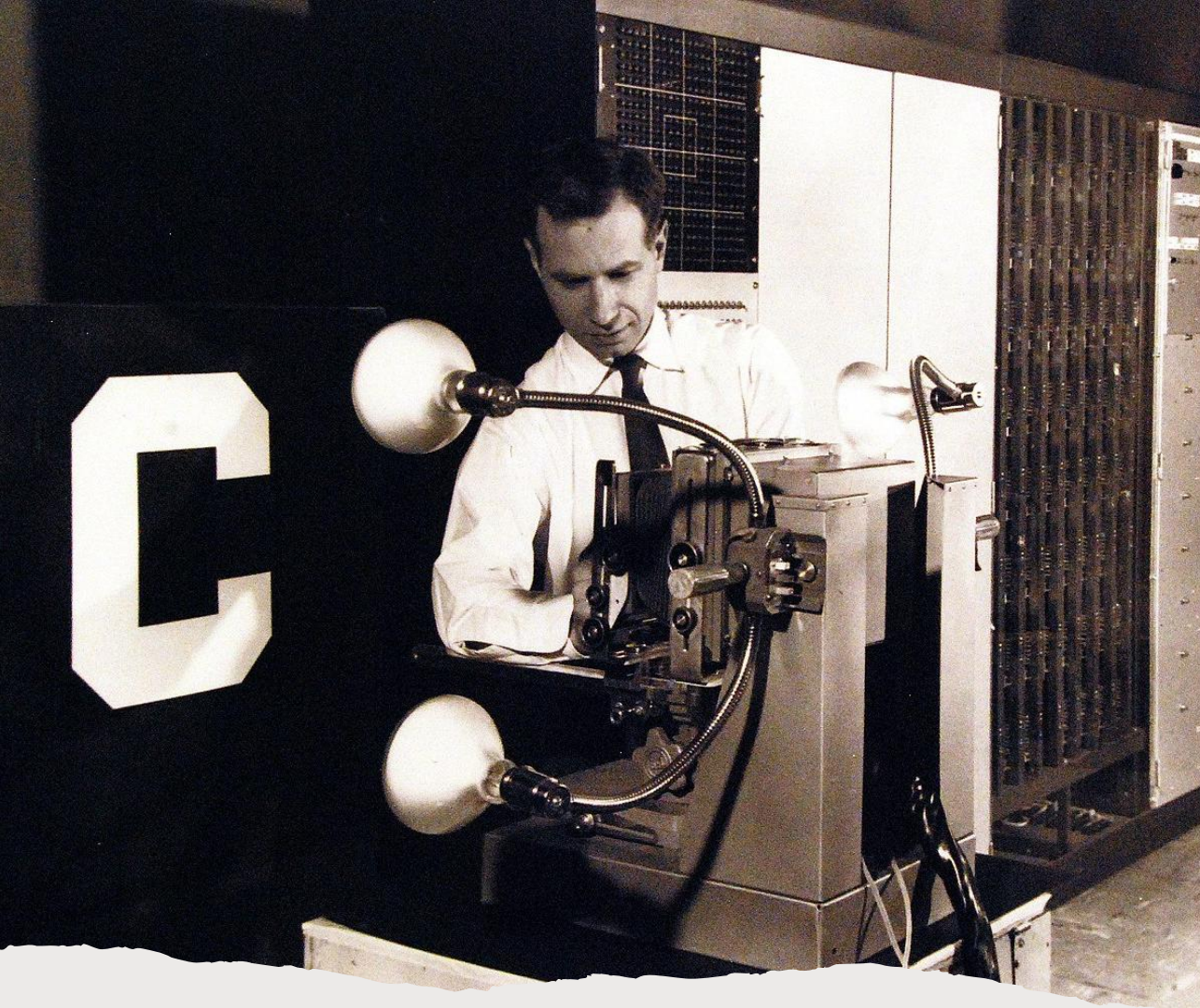
Toon Calders

Universiteit
Antwerpen





1950: Turing Test



1956: Perceptron

1960 - ELIZA

Talk to Eliza by typing your questions and answers in the input box.

> Hello, I am Eliza. I'll be your therapist today.

TYPE HERE 

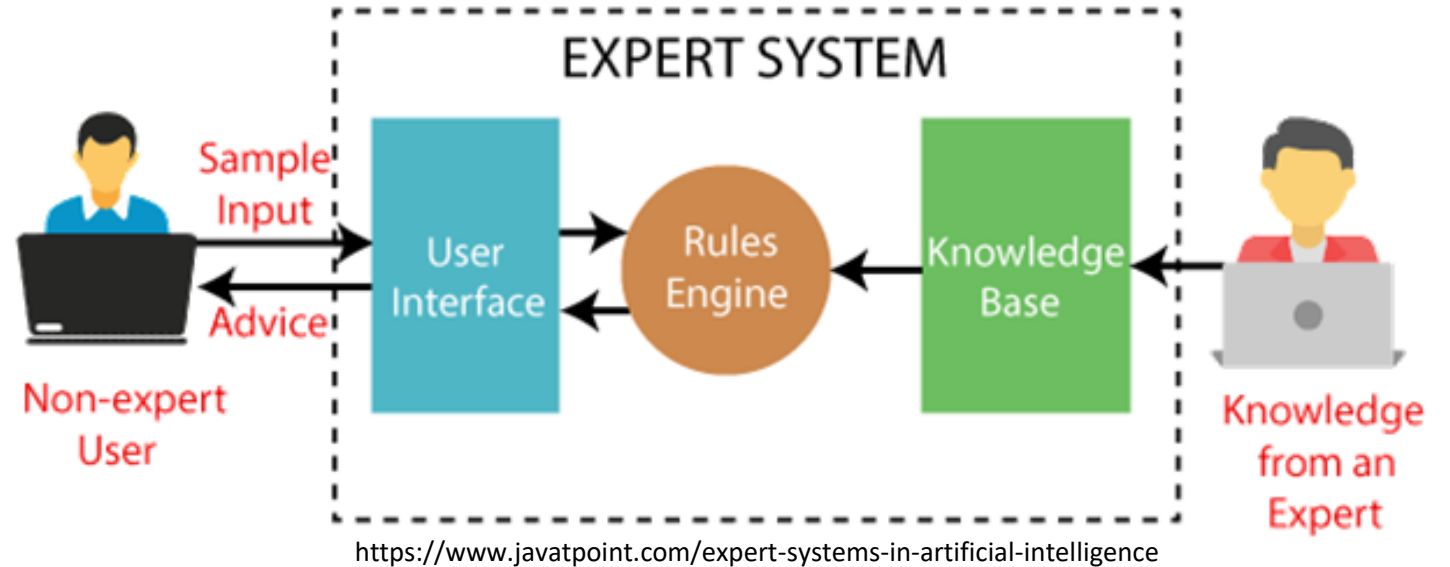
RECORDED WITH SCREENCAST  MATIC



70's AI Winter



80's Expert Systems



IF the infection is primary-bacteremia
AND the site of the culture is one of the sterile sites
AND the suspected portal of entry is the gastrointestinal tract
THEN there is suggestive evidence (0.7) that infection is bacteroid.

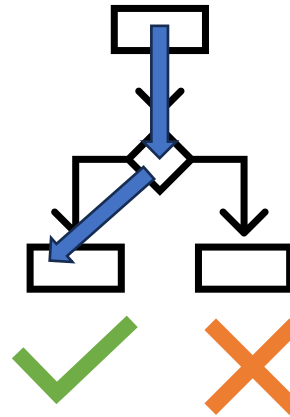
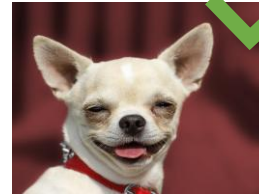
90's Machine Learning



Supervised learning

Reinforcement learning

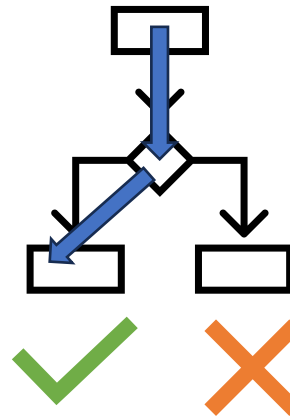
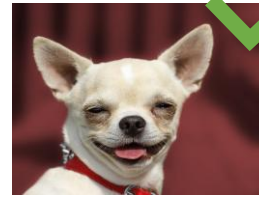
90's Machine Learning



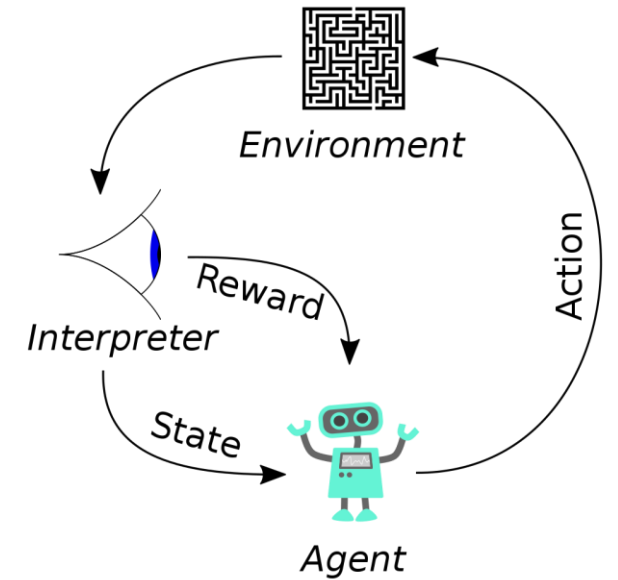
Supervised learning

Reinforcement learning

90's Machine Learning



Supervised learning



Reinforcement learning



Deep Blue (1997)



Watson (2011)

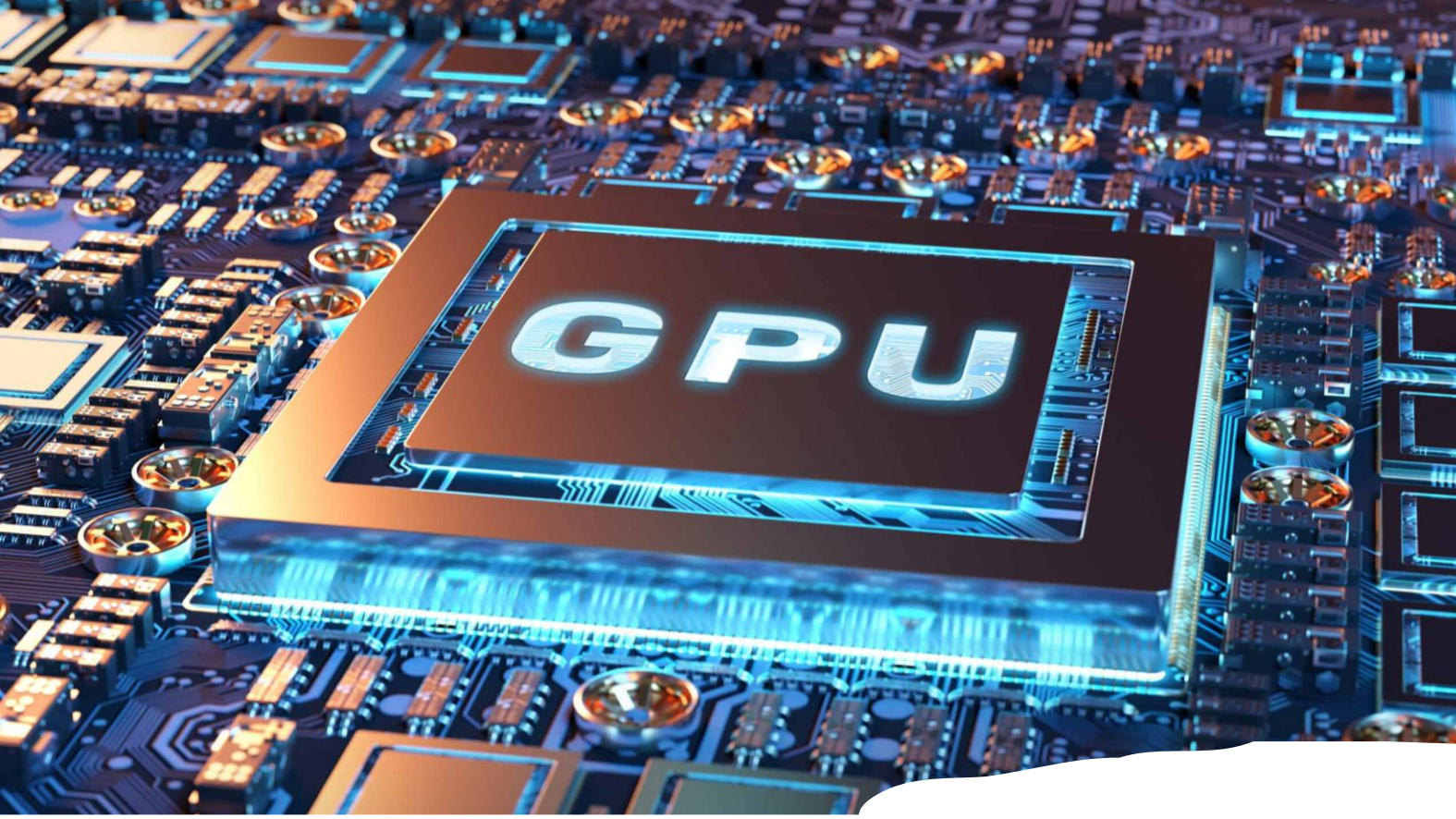


AlphaGo (2016)

Tom Mihalek/AFP/Getty Images

<https://en.wikipedia.org/w/index.php?curid=31142331>

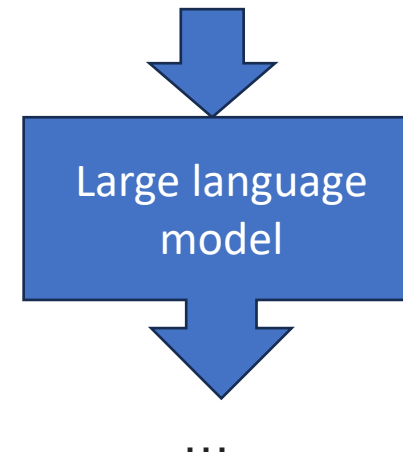
[https://medium.com/point-nine-news/
what-does-alphago-vs-8dadec65aaf](https://medium.com/point-nine-news/what-does-alphago-vs-8dadec65aaf)



What changed?

2020s : Generative AI

Generative artificial intelligence
is artificial intelligence capable of
generating text, images **and**



Pre-training

2020s : Generative AI



You

Give a one paragraph summary for a talk on generative AI



ChatGPT

Generative AI represents a groundbreaking leap in artificial intelligence technology, empowering machines to create new, original content by learning from vast datasets. This technology spans across various domains, including text generation, image creation, music composition, and more, revolutionizing how we interact with and conceive digital content. Through advanced algorithms like deep learning and neural networks, generative AI systems can mimic human creativity, producing work that ranges from realistic images to compelling narratives. This talk will explore the principles underlying generative AI, its current applications, and the ethical considerations it raises, as well as speculate on its future impacts on industries such as entertainment, art, and journalism. The discussion will highlight both the potential of generative AI to augment human creativity and the challenges it poses in terms of copyright, authenticity, and the changing landscape of creative work.

< 2 / 2 >    

Was this response better or worse?


Better

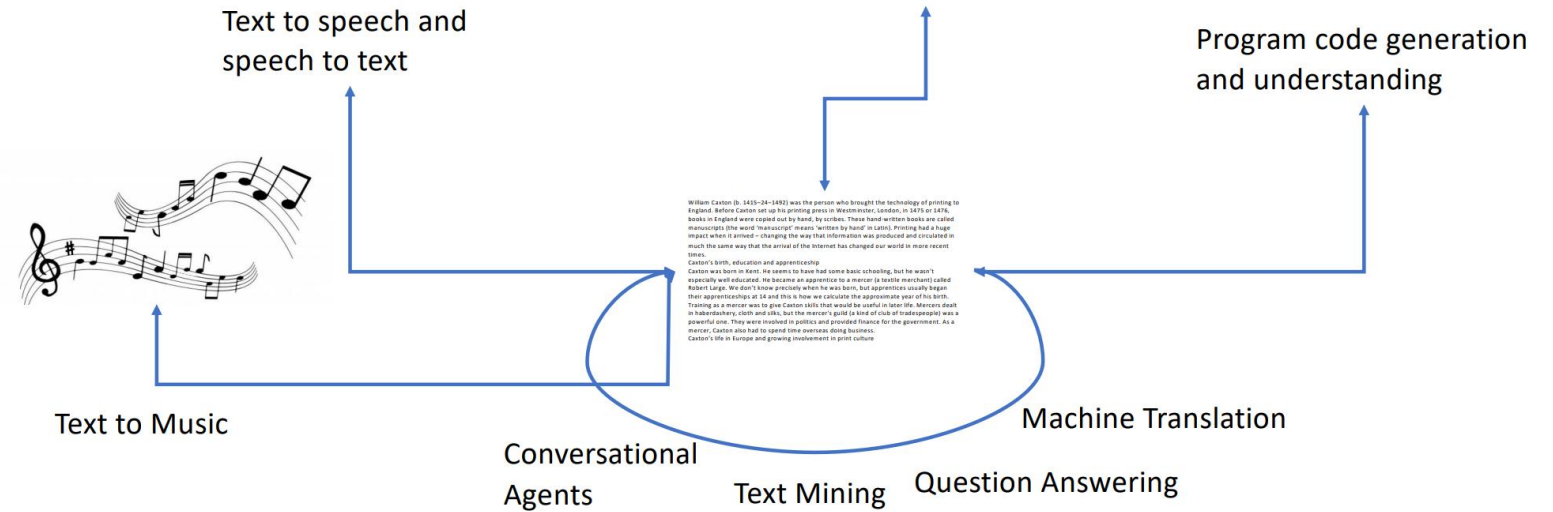

Worse


Same

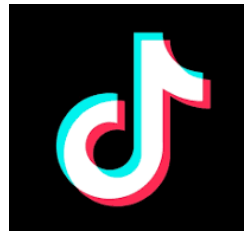


Reinforcement learning with human feedback

2020s : Multi-modal AI



You are a
Daily User of
AI



1960
Haha, loser!



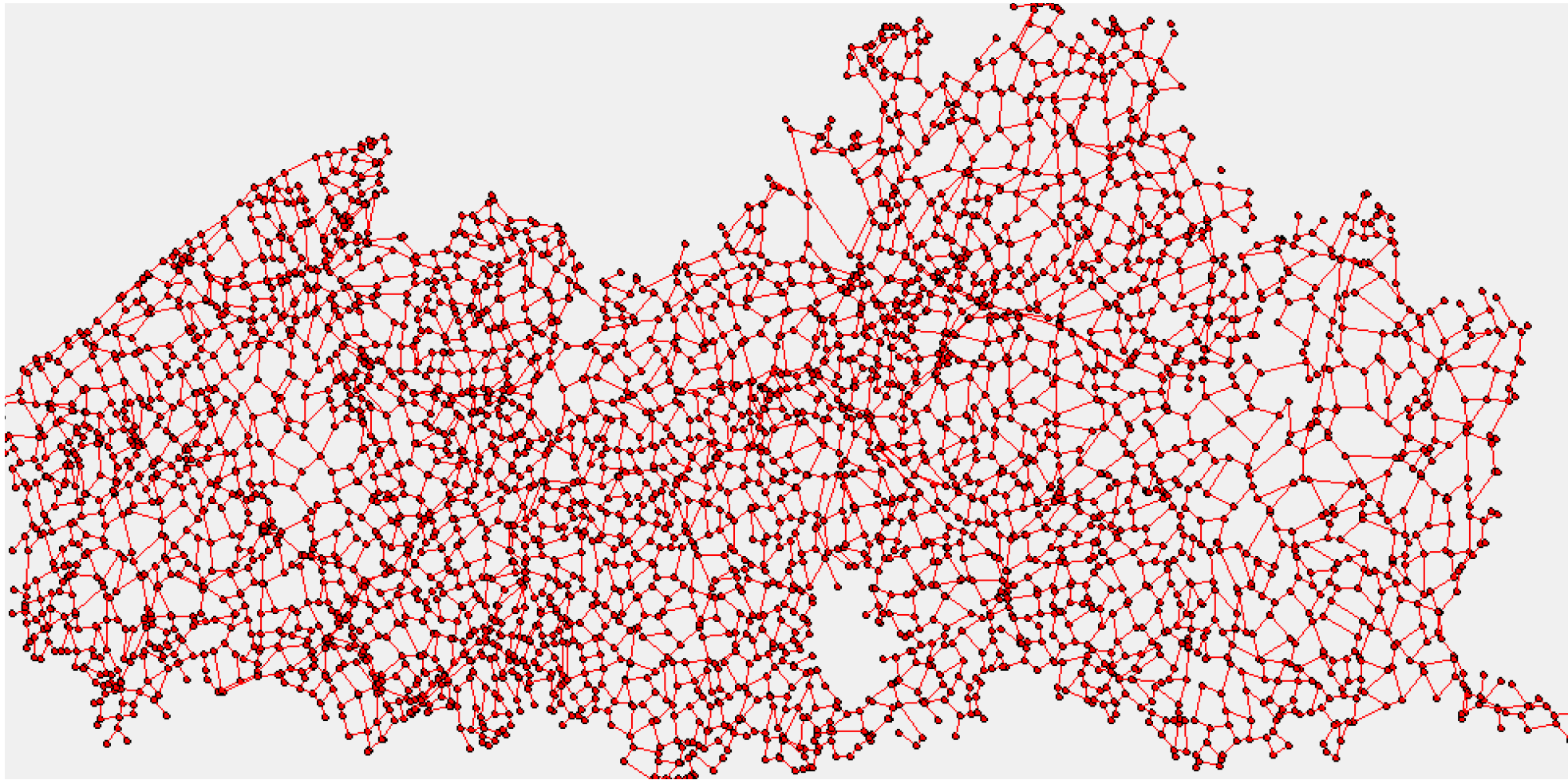
2024
Show off ...

Course Organization

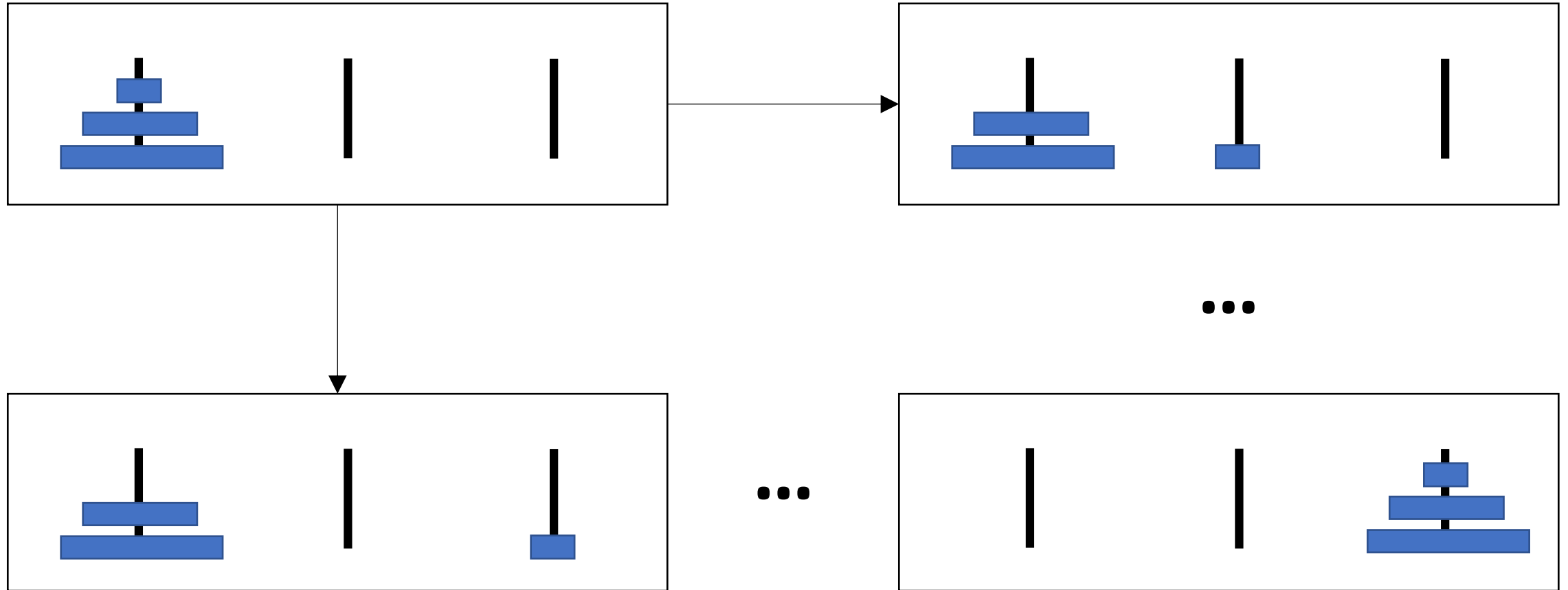
Course Content

- Lectures
 - Introduction (this lecture)
 - 12 theory lectures
 - Some practice sessions (solving exam questions of previous years)
- Project (20%)
 - First part is individual
 - Second part in small groups
 - All with oral defence
- Exam (80%)
 - Closed book; combination of theory and exercises

Course Topic 1&2: Search in Graphs

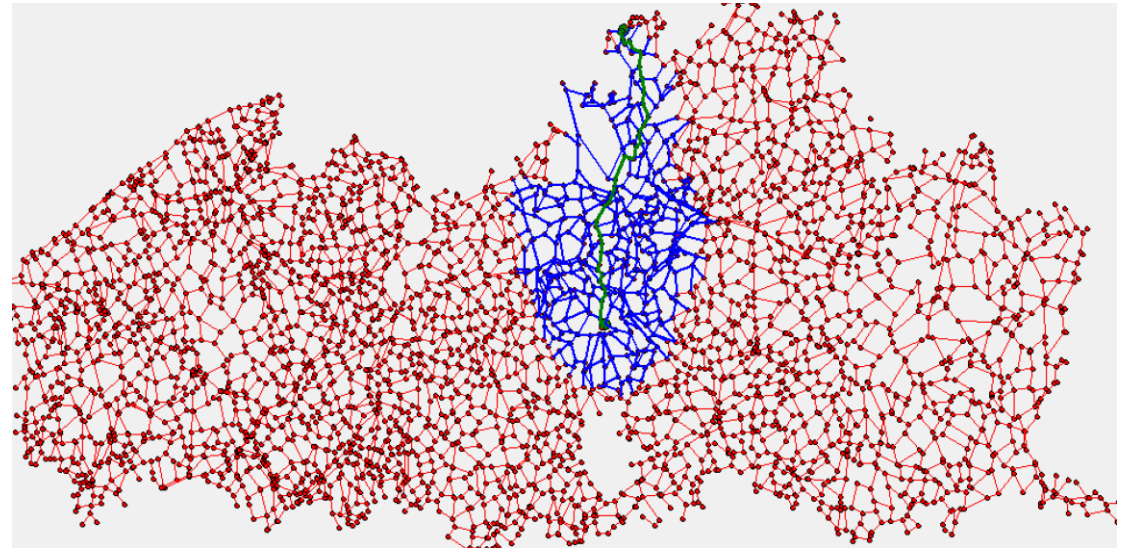
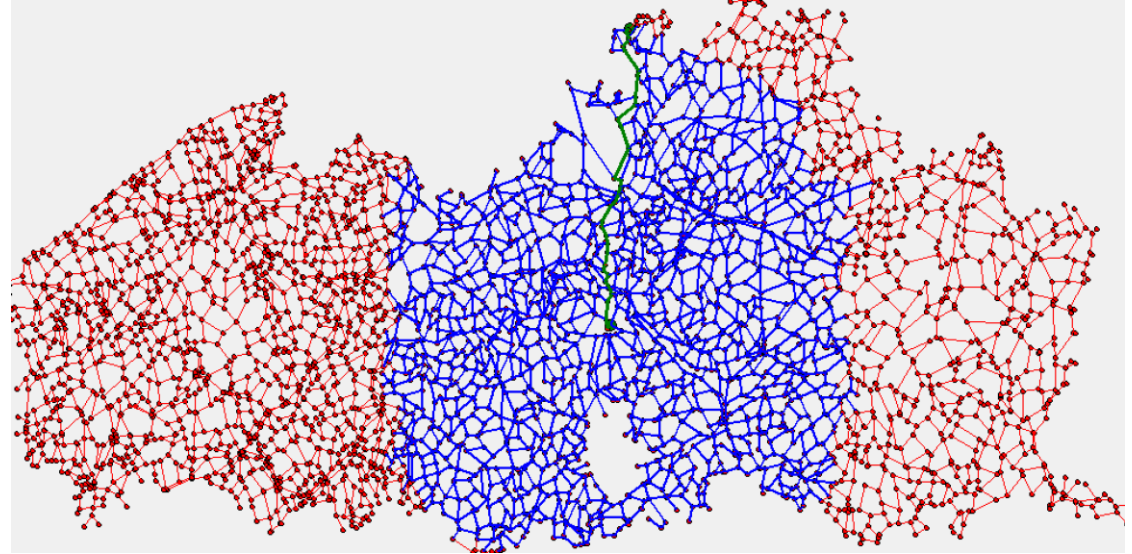


Course Topic 1&2: Search in Graphs



Course Topic 1&2: Search in Graphs

- Uninformed Search
- Informed Search

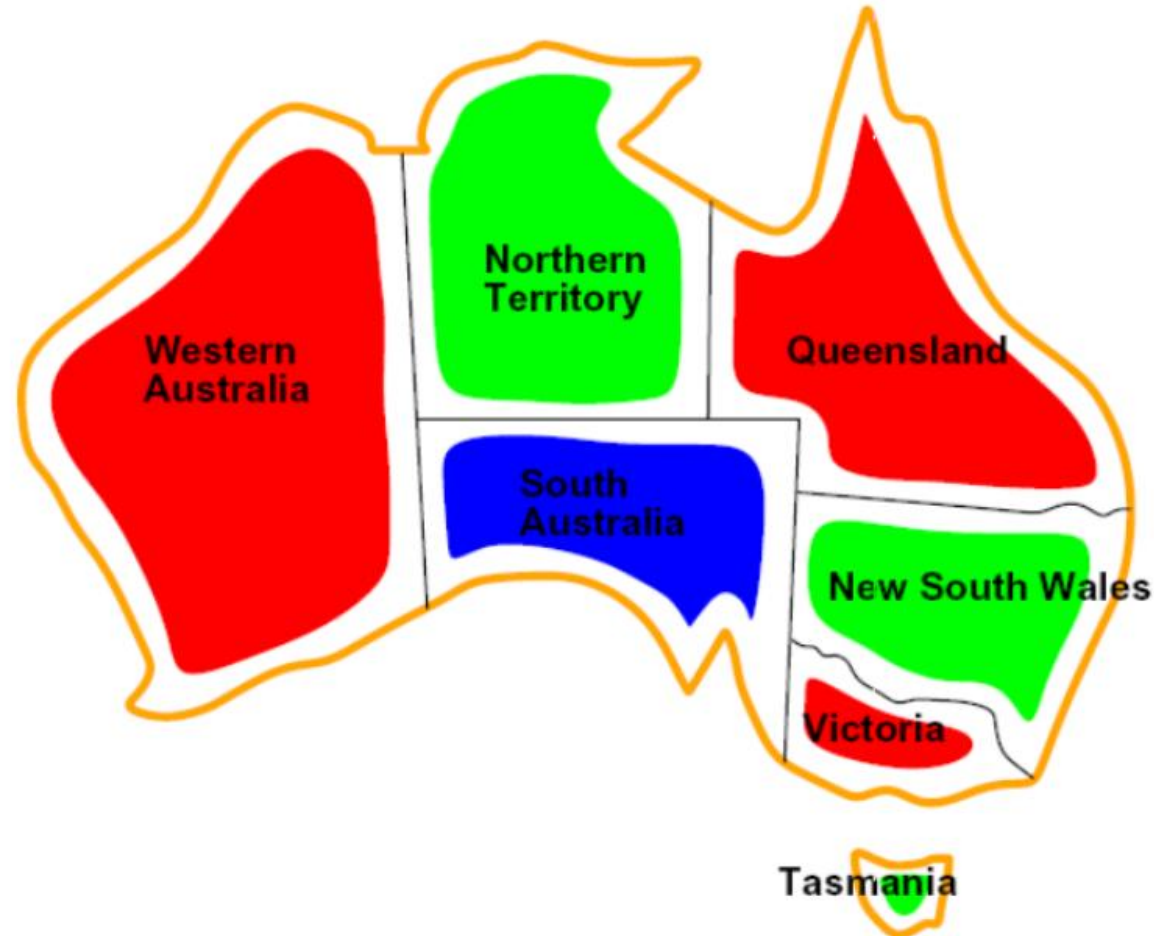


Course Topic 3: Adversarial Search

- Search, but with an opponent

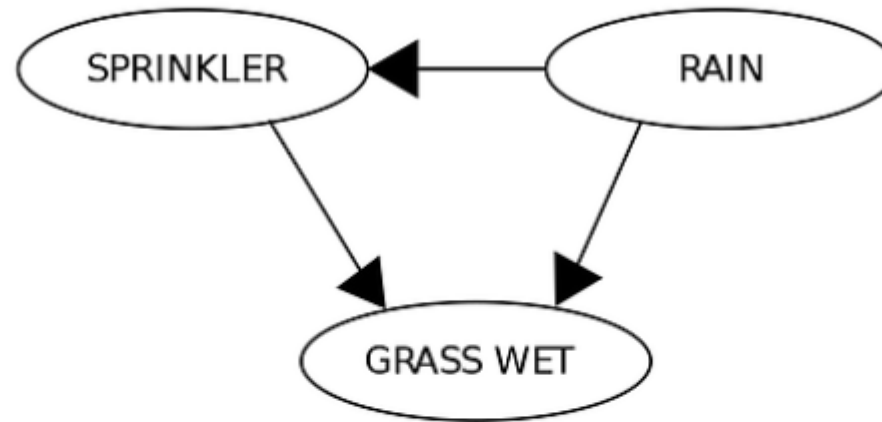


Course Topic 4: Constraint Satisfaction Problems



Course Topic 5: Bayesian Networks

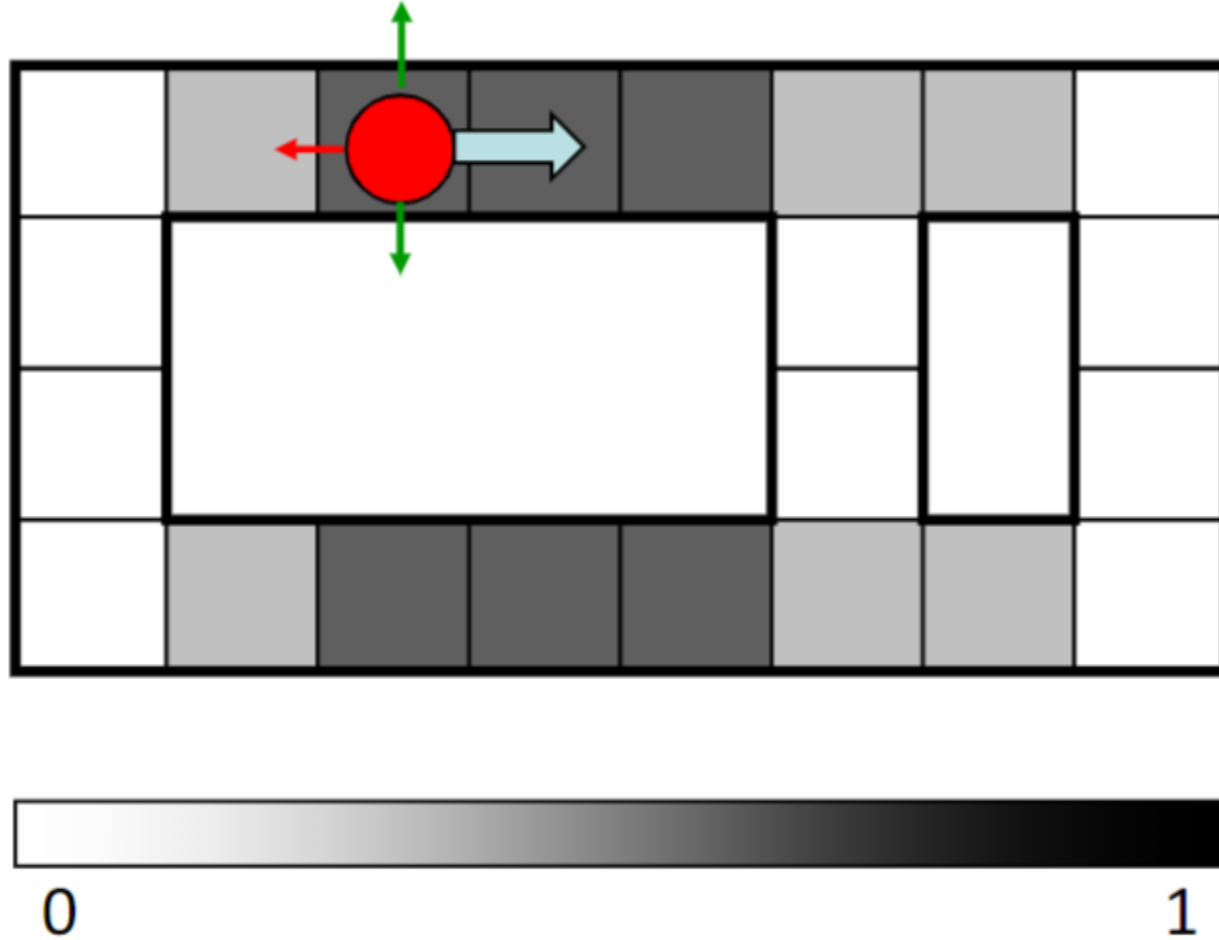
RAIN	SPRINKLER	
	T	F
F	0.4	0.6
T	0.01	0.99



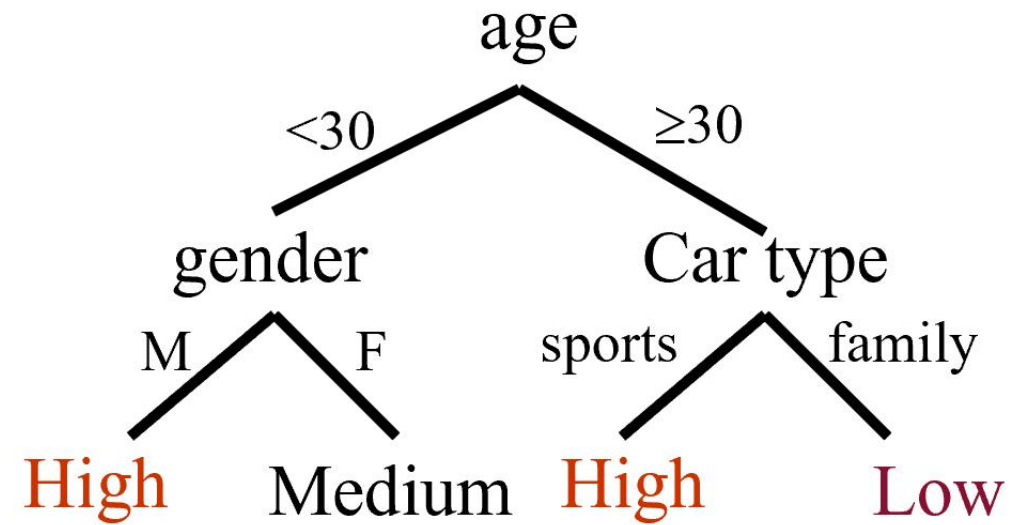
	RAIN	
	T	F
	0.2	0.8

SPRINKLER	RAIN	GRASS WET	
		T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

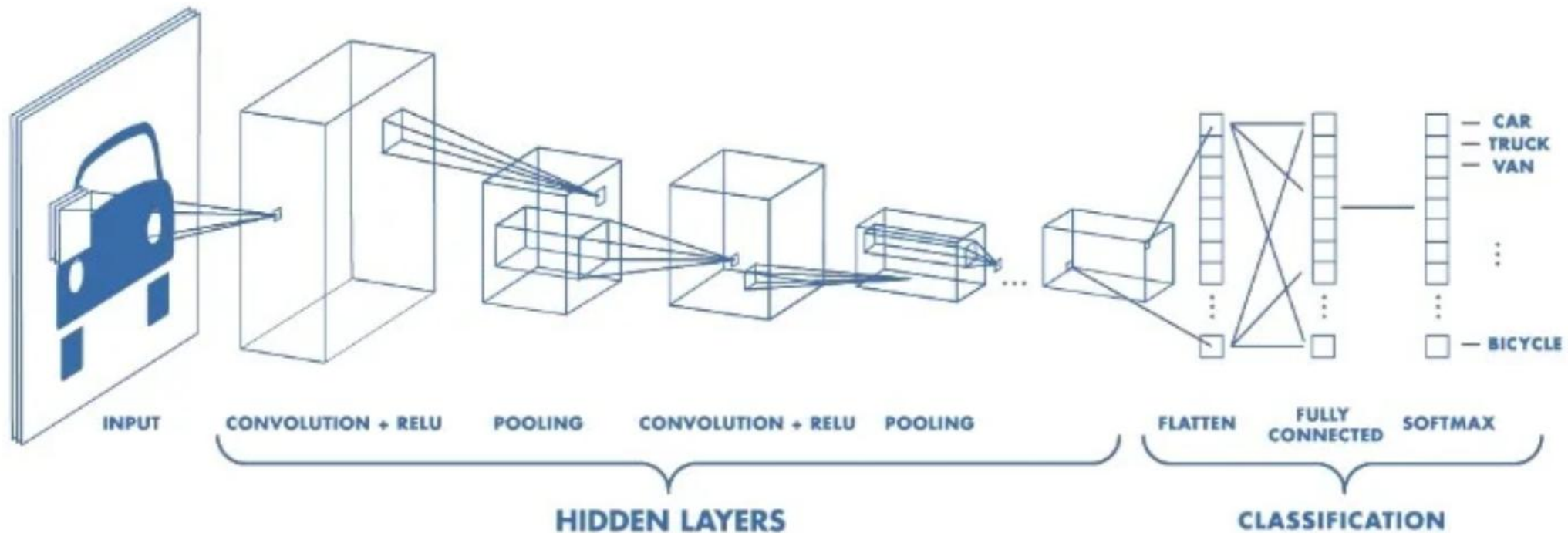
Course Topic 6: Markov Models



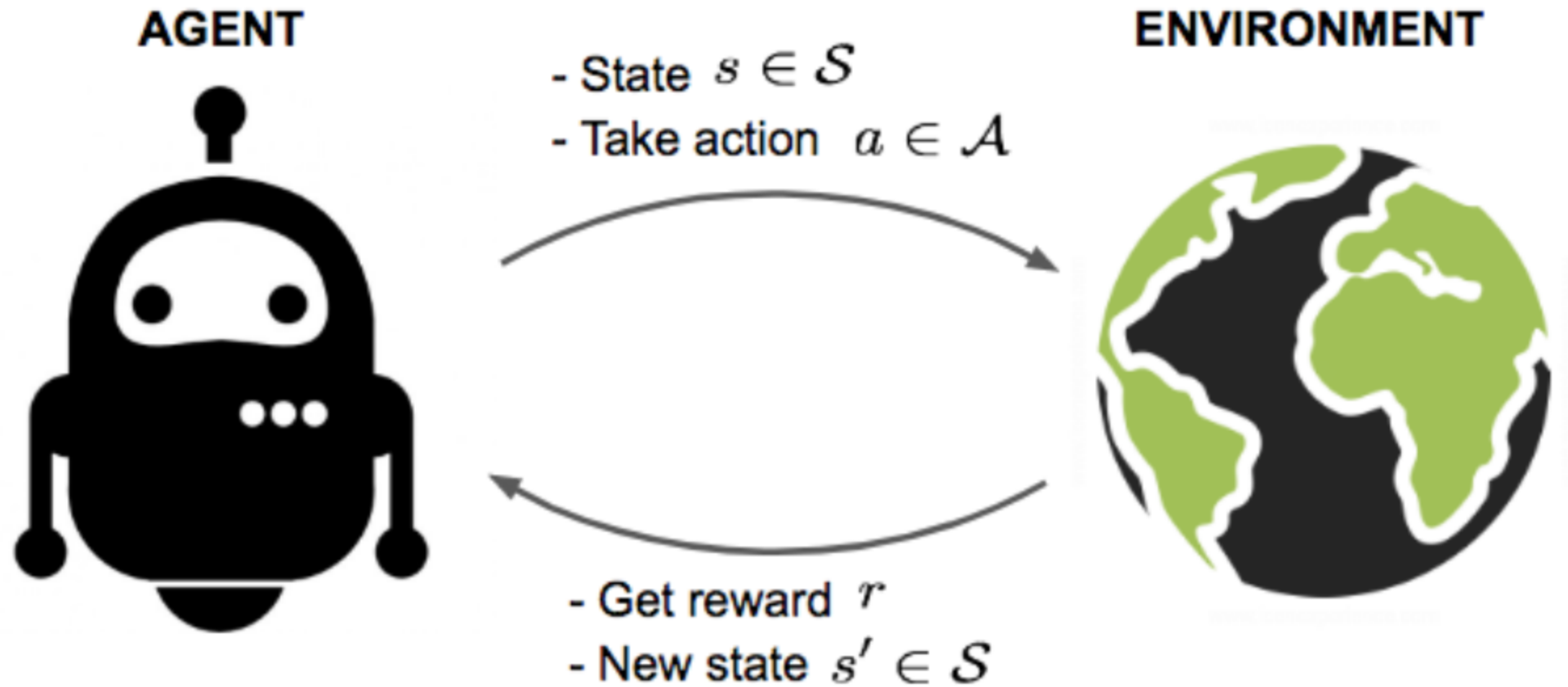
Course Topic 7: Machine Learning



Course Topic 8: Neural Networks



Course Topic 9: Reinforcement Learning



Summary

- Historical overview of AI
 - Search
 - Adversarial search
 - Uncertainty
 - Supervised, unsupervised, reinforcement learning
 - Deep neural networks
- Project
 - 4/20 points
 - No resit possible

Planning

29-9-2026	08:30	M.G.005		Les 1 - inleiding + uninformed search	
1-10-2026	13:45	M.G.026		Les 2 - informed search	
6-10-2026	08:30	M.G.005		Les 3 - adversarial search	
8-10-2026	13:45	M.G.006		Les 4 - constraint satisfaction problems	
13-10-2026	08:30	M.G.005		Practicum 1 - theoretische examenvragen search + CSP	
15-10-2026	13:45	M.G.026		Practicum 2 - praktijkoefeningen op computer	OR-tools
20-10-2026	08:30	M.G.005		Les 5 - Bayesian networks 1	
22-10-2026	13:45	M.G.006		Les 6 - Bayesian networks 2	
27-10-2026	08:30	M.G.005		Les 7 - Hidden Markov Models	
29-10-2026	13:45	M.G.026		Practicum 3 - Examenvragen BM + HMM	Vrijgeven project

Planning

3-11-2026	08:30	M.G.005		Les 8 - Machine learning - supervised learning : KNN, Naive Bayes, Decision trees	
5-11-2026	13:45	M.G.026			
10-11-2026	08:30	M.G.005		Les 9 - ANN	
12-11-2026	13:45			Practicum 4: ML en ANN	
17-11-2026	08:30	M.G.005		Les 10 - afwerken deep learning; start MDPs	
19-11-2026	13:45	M.G.026		Practicum 5 - Project verdediging deel 1 + Punten deel 1 met Poll	
24-11-2026	08:30	M.G.005		Les 11 - Afwerken MDPs, start Reinforcement learning	
26-11-2026	13:45	M.G.026		Practicum 6 - Feedback moment	
1-12-2026	08:30	M.G.005		Les 12 - Afwerken Reinforcement learning en afronden	

Planning

3-12-2026	13:45			
8-12-2026	08:30			
10-12-2026	13:45			
15-12-2026	08:30	M.G.005		Practicum 7 - Mondelinge verdediging project
17-12-2026	13:45	M.G.026		Practicum 7 - Mondelinge verdediging project

